

AI-POWERED STUDENT COURSE ALLOCATION SYSTEM

API Documentation

Base URL

http://localhost:5000/api

Authentication APIs

1. Login

POST /auth/login

Request

```
{  
  "username": "admin",  
  "password": "password123"  
}
```

Success Response

```
{  
  "token": "jwt_token",  
  "user": {  
    "id": 1,  
    "username": "admin",  
    "role": "admin"  
  }  
}
```

2. Verify Authentication

GET /auth/verify

Headers

Authorization: Bearer <JWT_TOKEN>

Success Response

```
{  
  "authenticated": true,  
  "user": {
```

```
"id": 1,  
"username": "admin",  
"role": "admin"  
}  
}
```

3. Change Password

PUT /auth/change-password

Headers

Authorization: Bearer <JWT_TOKEN>

Request

```
{  
  "currentPassword": "oldPassword",  
  "newPassword": "newPassword"  
}
```

Student APIs

Get All Students

GET /students

Get Student by ID

GET /students/{id}

Example

GET /students/1

Create Student

POST /students

Example Request

```
{  
  "name": "John",  
  "email": "john@example.com",
```

```
"department": "Computer Science"
}
```

Update Student

PUT /students/{id}

Delete Student

DELETE /students/{id}

Course APIs

Get All Courses

GET /courses

Get Course by ID

GET /courses/{id}

Create Course

POST /courses

Example

```
{
  "courseName": "Data Structures",
  "faculty": "Dr. Kumar",
  "capacity": 60
}
```

Update Course

PUT /courses/{id}

Delete Course

DELETE /courses/{id}

Course Allocation APIs

Run Allocation

POST /allocations/run

Runs the course allocation algorithm and assigns students to available courses.

Get All Allocations

GET /allocations

Returns all allocated students and courses.

Allocation Statistics

GET /allocations/stats

Returns allocation summary and statistics.

Reset Allocations

DELETE /allocations/reset

Removes all existing allocations from the system.

AI APIs

AI Query

POST /ai/query

Request

```
{  
  "prompt": "Suggest a suitable course for a student interested in Artificial Intelligence."  
}
```

Response

```
{  
  "response": "Recommended course based on the provided prompt..."  
}
```

Authentication

Protected APIs require a JWT token.

Example Header

Authorization: Bearer <JWT_TOKEN>
